Data Oversampling & Synthesis Specification

Methodology, Distribution Assumptions, and Generation Statistics

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1. Purpose & Analytical Motivation

To simulate a realistic enterprise cold chain environment without altering historical records, an oversampling mechanism was designed. In a production pharmaceutical setting, historical operational records up to 2025-12-31 must remain immutable, while newly generated streaming batches reflect subsequent operational cycles into 2026. This allows verifying pipeline idempotency, incremental partition handling, and dashboard refreshes.

2. Generation Logic & Referential Integrity

The synthesis engine (`scripts/oversample.py`) executes the following rules:

- **Primary Key Monotonicity:** The generator queries existing primary keys across both historical (`data/raw/`) and landing directories (`data/landing/`). New IDs resume strictly above the maximum observed value (e.g., sales transaction IDs continue sequentially from `SALE-000500001`).
- **Foreign Key Validation:** Generated rows only reference valid `MedicineID` foreign keys sourced directly from `01_master_data.csv` (63 SKU catalogue), preventing orphan facts.
- **Temporal Forwarding:** Synthetic transaction timestamps are pushed forward into 2026 (starting from `2026-01-01`), enabling clear temporal segmentation between historical baselines and newly simulated intake.
- **Hospital Grid Completeness:** Hospital demand batches generate exactly 756 records per cycle (63 medicines × 12 regional hospitals = 1 full month of demand across the network).

3. Generation Volume & Frequency Rationale

The DAG synthesizes two incremental batches per pipeline execution: 10,000 sales transactions and 756 hospital demand records per batch. This volume was selected to simulate meaningful statistical variance during dashboard refreshes while avoiding pipeline memory saturation within the containerized local environment.

4. Quantitative Synthesis Statistics

Dataset	Historical Baseline	Batch Increment	Staging Total (Post-Run)	Key Generation Format
Pharmacy Sales	500,000 rows	10,000 rows × 2 batches	520,000 rows	`SALE-%09d` (sequential)
Hospital Demand	27,216 rows	756 rows × 2 batches	28,728 rows	`HOSP-DEM-%08d`
Master Catalogue	63 SKUs	0 (SCD Type 2 maintained)	64 dimensional rows	`MED-0001` to `MED-0063`

> **Core Assumption:** Synthetic oversampling maintains the underlying statistical and price variance of the original historical dataset. Seasonal epidemiological spikes and supply disruptions are intentionally omitted for Phase 1 to preserve predictable baseline behavior.